

records, it is doubtful whether they would have any real utility for the future. In life insurance the relation between age and mortality rate is well defined and changes only gradually. The same is true, to a much lesser extent, of the relation between the various types of structures and the fire hazard attaching to them. But the relation between different kinds of investments and the risk of loss is entirely too indefinite, and too variable with changing conditions, to permit of sound mathematical formulation. This is particularly true because investment losses are not distributed fairly evenly in point of time, but tend to be concentrated at intervals, *i.e.*, during periods of general depression. Hence the typical investment hazard is roughly similar to the conflagration or epidemic hazard, which is the exceptional and incalculable factor in fire or life insurance.

Self-Insurance Generally Not Possible in Investment. If we were to assume that a precise mathematical relationship does exist between yield and risk, then the result of this premise should be inevitably to recommend the lowest yielding—and therefore the safest—bonds to all investors. For the individual is not qualified to be an insurance underwriter. It is not his function to be paid for incurring risks; on the contrary it is to his interest to pay others for insurance against loss. Let us assume a bond buyer has his choice of investing \$1,000 for \$20 per annum without risk, or for \$70 per annum with 1 chance out of 20 each year that his principal would be lost. The \$50 additional income on the second investment is mathematically equivalent to the risk involved. But in terms of *personal requirements*, an investor cannot afford to take even a small chance of losing \$1,000 of principal in return for an extra \$50 of income. Such a procedure would be the direct opposite of the standard procedure of *paying* small annual sums to protect property values against loss by fire and theft.

The Factor of Cyclical Risks. The investor cannot prudently turn himself into an insurance company and incur risks of losing his principal in exchange for annual premiums in the form of extra-large interest coupons. One objection to such a policy is that sound insurance practice requires a very wide distribution of risk, in order to minimize the influence of luck and to allow maximum play to the law of probability. The investor may endeavor to attain this end by diversifying his holdings,